

# Tianyue H. Zhang

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Github: [tianyuehz](https://github.com/tianyuehz) · Publication: [Google Scholar](#)

## Education

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| Sept. 2023 - Present    | <b>Mila/Université de Montréal</b> GPA: $A^+ \equiv 4.33$<br><i>Ph.D. Computer Science, DIRO</i><br>Supervised by Dr. Simon Lacoste-Julien                                             |
| Sept. 2021 - Sept. 2023 | <b>University of British Columbia</b> GPA: $95.3/100 \equiv 4.33$<br><i>M.Sc. Computer Science</i><br>Supervised by Dr. Mark Schmidt                                                   |
| Sept. 2015 - Sept. 2019 | <b>University of British Columbia</b> GPA: $85/100 \equiv 3.95$<br><i>B.Sc. Combined Honours in Computer Science and Mathematics</i><br>Dean's Honour List; Graduated with Distinction |

## Publications

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**Deriving Hyperparameter Scaling Laws via Modern Optimization Theory** [Paper](#)

Sci4DL workshop at ICLR 2026

E. Shulgin, D. von Rütte, T. Zhang, N. Ajroldi, B. Schölkopf, A. Orvieto

**Understanding Adam Requires Better Rotation Dependent Assumptions** [Paper](#)

NeurIPS 2025

T. Zhang<sup>\*</sup>, L. Maes<sup>\*</sup>, A. Milligan, A. Jolicoeur-Martineau, I. Mitliagkas, D. Scieur, S. Lacoste-Julien, C. Guille-Escuret

**Addressing Concept Mislabeling in Concept Bottleneck Models Through Preference Optimization** [Paper](#)

ICML 2025; Bi-Align Workshop Workshop at ICLR 2025 (Oral)

E. Penalosa<sup>\*</sup>, T. Zhang, L. Charlin, M. E. Zarlenga

**Connecting Thompson Sampling and UCB: Towards More Efficient Trade-offs Between Privacy and Regret** [Paper](#)

ICML 2025

B. Hu, Z. Huang, T. Zhang, M. Lecuyer, N. Hegde

**On PI Controllers for Updating Lagrange Multipliers in Constrained Optimization** [Paper](#)

ICML 2024

M. Sohrabi<sup>\*</sup>, J. Ramirez<sup>\*</sup>, T. Zhang, S. Lacoste-Julien, J. Gallego-Posada

**Optimistic Thompson Sampling for Episodic Reinforcement Learning** [Paper](#)

UAI 2023

B. Hu, T. Zhang, N. Hegde, M. Schmidt

## Selected Awards

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|      |                                                                |
|------|----------------------------------------------------------------|
| 2025 | NSERC PGSD, \$40000 annually                                   |
| 2025 | Mila Excellence Scholarships – Women in AI, \$8000 annually    |
| 2023 | Professor Cho Diversity Scholarship, \$1500                    |
| 2022 | UBC Cloud Innovation Center Fellowship, \$20000                |
| 2019 | NSERC Undergraduate Student Research Award, \$6000             |
| 2018 | UBC Science Faculty International Student Scholarship, \$10000 |
| 2015 | UBC Outstanding International Student Award, \$6000            |

## Thesis

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**Optimistic Thompson Sampling: Strategic Exploration in Bandits and RL** [Thesis](#)

Supervisory committee: Dr. Mark Schmidt and Dr. Mathias Lecuyer

## Selected Service

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|------------------|------------------------------------------------------|
| Reviewer         | ICML 2025; ICLR 2025; NeurIPS 2024                   |
| Organizer        | <a href="#">Montreal MLOpt</a>                       |
| Organizer        | <a href="#">Women@Mila</a>                           |
| Volunteer        | WiML; NeurIPS 2023; UAI 2023                         |
| Student Reviewer | Graduate Recruiting Committee at Mila 2024; UBC 2022 |
| Volunteer        | UBC Computer Science Tri-Mentoring Program, 2018     |

## Teaching Assistant

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|------------------------|-------------------------------------------------------|
| Sept. 2023 - Dec. 2023 | Polytechnique Montreal INF8245E - Machine Learning    |
| Sept. 2021 - May 2023  | UBC · CPSC 532M/340: Machine Learning and Data Mining |
| Jan. 2018 - Apr. 2018  | UBC · Math 152: Linear Systems                        |
| Sept. 2017 - Apr. 2018 | UBC · CPSC 121: Models of Computation                 |
| July 2017 - Aug. 2017  | UBC · Math 102: Integral Calculus                     |

## Selected Presentations

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| Nov. 2024 | <b>Understanding Adam Requires Better Rotation Dependent Assumptions</b><br><i>MTL MLOpt</i>                                                                                   |
| June 2024 | <b>On PI Controllers for Updating Lagrange Multipliers in Constrained Optimization</b><br><i>Tsinghua University, Statistical Artificial Intelligence &amp; Learning Group</i> |
| Apr. 2023 | <b>Deep exploration via randomized value function</b><br><i>UBC · Reinforcement Learning Reading Group</i>                                                                     |
| Oct. 2022 | <b>Language Models are Few Shot Learners (GPT-3)</b><br><i>UBC · Machine Learning Reading Group</i>                                                                            |
| Dec. 2021 | <b>Reinforcement Learning and Autonomous Driving</b><br><i>UBC · CPSC 340: Machine Learning and Data Mining</i>                                                                |
| Nov. 2021 | <b>Probabilistic Topic Modeling</b><br><i>UBC · CPSC 503: Computational Linguistics</i>                                                                                        |

## Work Experience

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| May 2022 - Dec. 2022   | <b>Research Mentor</b><br><i>UBC-AWS Cloud Innovation Center</i><br>Advised undergraduate students in building machine learning model prototypes in collaboration with AWS                                              |
| Sept. 2019 - Apr. 2021 | <b>Software Engineer</b><br><i>Magnitude - Simba Technologies</i><br>Developed JDBC driver for Amazon Redshift Database using Java and SQL;<br>Designed test packages and website interface using React.js and HTML/CSS |
| July 2018 - Aug. 2018  | <b>Research Intern</b><br><i>Bank of China International, London</i><br>Trained regression models for U.S. 10 Year Treasury data and composed financial reports on futures and options predictions                      |

## Competitions

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|------------|-----------------------------------------------------------------------------------------------------------|
| Dec. 2018  | <b>Putnam Competition, Mathematical Association of America</b><br><i>Ranked top 1500 in North America</i> |
| Sept. 2014 | <b>Chinese Physics Olympiad (CPhO)</b><br><i>Guangdong Provincial Secondary Prize</i>                     |

## Language

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Mandarin (native), English (fluent), Cantonese (intermediate), French (beginner)